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COS700 Research Proposal

Developing Multilingual South African Context Online Cybersecurity Modules for Students

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Abstract

Cybersecurity threats continue to increase as digital technologies and internet usage expand. Students are increasingly exposed to cyber risks such as phishing, scams, and social engineering attacks. Although many cybersecurity learning platforms exist, most are designed for general audiences and lack contextual relevance to specific regions, including South Africa.

This research aims to develop South African-context online cybersecurity modules that address cybersecurity threats relevant to students. The study will adopt a qualitative research approach, using a systematic literature review guided by the PRISMA framework to analyse existing cybersecurity education platforms, identify gaps, and inform the design of the proposed modules. The modules will be developed using an instructional design approach to ensure that the learning content is structured, engaging, and suitable for students. The modules will focus on interactive and engaging learning approaches by reducing text-heavy content and incorporating practical scenarios, AI-assisted learning content, and real-world examples that demonstrate how cyber threats occur, how attackers disguise malicious activities, and how users can respond appropriately.

Additionally, the study will explore multilingual accessibility by providing cybersecurity learning content in selected South African languages, including isiZulu, Sepedi, and Xitsonga, to improve understanding and accessibility among diverse learners. The expected outcome of this research is to provide engaging, accessible, and locally relevant cybersecurity education resources that improve students' awareness and ability to identify and respond to cyber threats.

Keywords

Multilingual, AI-assisted learning content, South African context, cybersecurity awareness, online modules, cybersecurity education

1 Introduction

Cybersecurity is the practice of protecting systems, networks, and data from unauthorised access, attacks, and damage while ensuring confidentiality, integrity, and availability of information. The rapid growth of digital technologies has increased dependence on online platforms for communication, education, banking, and transactions. This growth has also resulted in an increase in cyber threats such as phishing, scams, social engineering, and data breaches.

The increase in cyber threats highlights the importance of cybersecurity awareness, especially among students who are frequent users of digital platforms. Many users lack sufficient knowledge of how cyber attacks occur, how attackers manipulate individuals, and how to protect themselves. Although cybersecurity education platforms exist, many focus on general concepts and do not consider the specific challenges experienced in different regions.

In South Africa, students and digital users experience localised cyber threats such as WhatsApp scams, email scams, SMS scams, and online delivery scams. Additionally, regulations such as the Protection of Personal Information Act (POPIA) and the Cybercrimes Act introduce cybersecurity requirements specific to the country. Therefore, developing cybersecurity resources that include South African examples and regulations can make learning more relevant and easier for students to understand.

This research proposes the development of South African-context online cybersecurity modules that use interactive learning approaches instead of relying mainly on theoretical content. The modules will incorporate AI-assisted learning content, practical scenarios, and examples showing how cyber threats occur and how users can respond. The study will also explore multilingual accessibility by incorporating isiZulu, Sepedi and Xitsonga to improve understanding among diverse learners and contribute towards more accessible cybersecurity education.

2 Problem Statement

Existing online cybersecurity modules are often designed for a global audience and do not sufficiently address cybersecurity challenges within the South African context. This limits students' exposure to local threats, legislation, and real-world scenarios they encounter.

Research Questions

1. How can South African-context online cybersecurity modules be developed to improve cybersecurity awareness among students?
2. Does contextualising cybersecurity modules around local South African scams improve students' understanding?
3. How can interactive learning approaches and AI-assisted content improve student engagement with cybersecurity education?

Main Deliverables

- Development of focused cybersecurity learning modules addressing relevant threats in South Africa.
- Focus on POPIA and digital privacy awareness, and South African online scams (WhatsApp, SMS, and delivery scams).
- AI-assisted scenarios demonstrating how cyber threats occur.
- Adaptation of cybersecurity content into selected South African languages: isiZulu, Sepedi and Xitsonga.
- Practical learning activities and assessments.
- A badge or certificate issued to students who pass the assessments.

3 Literature Study

Cybersecurity education has become increasingly important due to the rapid growth of digital technologies and the rise in cyber threats such as phishing, social engineering, scams, and data breaches (Modise 2025). Existing cybersecurity learning platforms aim to improve awareness and digital safety; however, many resources provide general cybersecurity knowledge without considering the specific needs and challenges of learners in different regions (Veerasamy et al. 2025). Research suggests that contextualised learning approaches improve understanding by connecting educational content to real-world experiences (da Veiga et al. 2022).

In the South African context, there is a need for cybersecurity education resources that address local threats and regulations. South African users experience cyber risks such as WhatsApp scams, email phishing, SMS scams, and online fraud, while laws such as the Protection of Personal Information Act (POPIA) and the Cybercrimes Act introduce country-specific cybersecurity considerations (Modise 2025). South Africa ranks among the lowest countries in the world for cyber risk literacy, with particular challenges around low digital literacy, multilingualism, and unequal access to technology (da Veiga et al. 2022). Additionally, many existing resources are mainly English-based and rely heavily on theoretical content, creating a need for more accessible, interactive, and multilingual learning approaches (Veerasamy et al. 2025). Von Solms & Van Heerden (Von Solms & Van Heerden 2022) further note that South African students often graduate from undergraduate programmes without sufficient practical cybersecurity knowledge, reinforcing the need for targeted, context-sensitive educational interventions.

Studies on digital learning highlight that interactive methods such as videos, scenarios, and practical activities can improve student engagement and knowledge retention (Luo 2024). Gamification elements such as digital badges have been shown to significantly enhance learners' intrinsic motivation across multiple dimensions, including interest, perceived competence, and the value placed on learning activities (Luo 2024). Furthermore, multilingual learning approaches can support better understanding among diverse learners, a consideration that is especially significant in a country with eleven official languages (Veerasamy

et al. 2025). Therefore, there is an opportunity to develop South African-context online cybersecurity modules that combine local cyber threats, interactive learning approaches, and support for languages such as isiZulu, Sepedi, and Xitsonga to improve cybersecurity awareness among students.

4 Methodology

Research Paradigm and Approach

The study will follow a Design Science Research (DSR) approach because the research aims to develop and evaluate an artefact, which in this case is South African-context online cybersecurity modules. This approach is appropriate because the research is focused on solving a practical problem by designing, implementing, and evaluating a cybersecurity education solution rather than only studying the problem.

Literature Review (PRISMA Approach)

A Systematic Literature Review (SLR) guided by the PRISMA framework will be conducted. The purpose is to identify existing cybersecurity education approaches, online learning modules, and gaps related to:

- South African contextualisation
- Interactive learning
- Multilingual accessibility
- Student cybersecurity awareness

Literature will be collected from academic databases and screened using PRISMA selection criteria.

Data Collection and Analysis

Data will be collected from:

- Research papers
- Existing cybersecurity education platforms
- Online learning resources

The collected information will be analysed to identify:

- Common cybersecurity topics
- Learning approaches used
- Limitations in existing resources
- Requirements for the proposed modules

Module Design and Development

Based on findings from the literature review, cybersecurity modules will be developed focusing on selected South African-relevant topics such as:

- POPIA and digital privacy
- Online scams (WhatsApp, SMS, and email scams)

The modules will incorporate:

- Interactive scenarios
- AI-assisted learning content and videos
- Multilingual support (isiZulu, Sepedi, and Xitsonga)

Expected Results

- Students who engage with the South African-context cybersecurity modules demonstrate improved awareness and understanding of local cyber threats such as WhatsApp scams, email scams, SMS scams, and online fraud.
- The interactive learning approach, including AI-assisted videos, scenarios, and practical activities, improves students' engagement and understanding of cybersecurity concepts compared to traditional theory-based content.
- The developed modules successfully incorporate South African cybersecurity context, including local examples, POPIA, and relevant cyber safety practices.
- The inclusion of multilingual support (isiZulu, Sepedi and Xitsonga) improves accessibility and understanding of cybersecurity concepts among diverse learners.
- The developed online modules are usable, relevant, and accessible to students with different levels of cybersecurity knowledge.

5 Planning

The table below outlines the project plan, including the tasks, start and end dates, and a brief description of each activity.

Task	Start Date	End Date	Description
Research Planning and Proposal Refinement	28 Mar 2026	27 May 2026	Understanding the research topic, refining the research problem, scope, and deliverables.

Task	Start Date	End Date	Description
Literature Review	28 May 2026	15 Jul 2026	Conducting a systematic literature review using the PRISMA framework to identify existing cybersecurity learning approaches, gaps, and requirements for SA-context cybersecurity learning.
Analysis of Existing Modules	16 May 2026	23 May 2026	Review existing cybersecurity learning platforms to analyse their content, teaching approaches, limitations, and lack of South African contextualisation and multilingual support.
Module Planning and Design	16 Jul 2026	30 Jul 2026	Define module structure, learning outcomes, topics, and activities. Focus on selected modules such as POPIA/digital privacy and South African online scams.
Development of Learning Content	31 Jul 2026	7 Aug 2026	Develop module content including explanations, practical scenarios, quizzes, scam examples, and AI-assisted videos/audio content to improve engagement.
Multilingual Content Development	8 Aug 2026	16 Aug 2026	Adapt selected cybersecurity content into isiZulu, Sepedi and Xitsonga to improve accessibility for diverse learners.
Prototype Development	17 Aug 2026	17 Sep 2026	Implement the cybersecurity modules on an online platform/LMS and integrate interactive features, multimedia content, and assessments.
Testing and Evaluation	18 Sep 2026	1 Oct 2026	Evaluate the developed modules.
Data Analysis and Module Improvement	2 Oct 2026	23 Oct 2026	Analyse evaluation feedback and improve the modules based on findings.
Final Presentation	29 Oct 2026	29 Oct 2026	Final presentation of the research project.
Final Report Submission	31 Oct 2026	31 Oct 2026	Submitting the final report.

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